MATHEMATICS / INDEPENDENT STUDY

MM7030

Category Theory

Complete solutions

Original teaching material aligned with Stockholm University course descriptions.
Independently authored; not an official university publication.

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How to use this volume E problems practice definitions and standard arguments; C problems combine ideas; A problems require synthesis or generalization. These are study difficulty labels, not official SU grading criteria. Time estimates are guides, not deadlines. Write a complete attempt before opening hints. In the hints volume, read Hint 1 first and return to the problem before consulting Hint 2. Solutions give a decisive idea, a complete argument, and criteria for self-assessment.

1 Unit 1 · Categories and duality

MM7030-U1.01 · E · 15 min
Learning objective: U1.1 Category axioms

Verify that nonnegative integers with an arrow $m \to n$ exactly when $m \leq n$ form a category.

Decisive idea. There is at most one arrow of each type.

Reflexivity supplies identities; transitivity supplies a composite whenever arrows are composable. Both possible associative composites have the same source and target and hence are the unique arrow of that type. Identity laws hold for the same reason.

Check your argument. Check identities, existence of composition, and associativity.

MM7030-U1.02 · E · 15 min
Learning objective: U1.1 One-object categories

Describe the one-object category associated with the additive monoid $\mathbb{N}$. What arrows are invertible?

Decisive idea. An inverse would be a nonnegative integer adding to zero.

There is one object, arrows are nonnegative integers, composition is addition, and the identity arrow is zero. Only zero is invertible, since $n + m = 0$ with both nonnegative forces both zero.

Check your argument. Distinguish the object from its many endomorphisms.

MM7030-U1.03 · E · 15 min
Learning objective: U1.2 Opposites

Describe the opposite of a poset category.

Decisive idea. An arrow $a \to b$ now means $b \leq a$.

The opposite has the same objects with the reversed order. Identities are unchanged, and transitivity of the reversed order supplies composition.

Check your argument. State the new arrow condition explicitly.

MM7030-U1.04 · E · 15 min
Learning objective: U1.1 Isomorphisms

Prove an isomorphism in the category of groups is exactly a bijective group homomorphism.

Decisive idea. Apply the original homomorphism to the desired inverse equation.

An inverse morphism makes the underlying function bijective. Conversely let $f : G \to H$ be a bijective homomorphism. For $x = f(a), y = f(b)$, $f^{-1}(xy) = f^{-1}(f(ab)) = ab = f^{-1}(x)f^{-1}(y)$. Thus its inverse function is a homomorphism.

Check your argument. Prove the reverse direction rather than treating inverse preservation as automatic.

MM7030-U1.05 · C · 30 min
Learning objective: U1.2 Monomorphisms
Prove the composite of two monomorphisms is monic.
Decisive idea. From $nmu = nmv$, first cancel n.
If n, m are monic and $(nm)u = (nm)v$, monicity of n gives $mu = mv$, then monicity of m gives $u = v$. This is the definition of monicity of the composite.
Check your argument. State the sources and targets needed for all composites.

MM7030-U1.06 · C · 30 min
Learning objective: U1.2 Cancellation
If gf is monic, prove f is monic. Must g be monic?
Decisive idea. For the counterexample use an inclusion followed by a constant map on a larger set.
If $fu = fv$, then $gfu = gfv$, so monicity of gf gives $u = v$. The second claim fails: let f include a singleton into a two-element set and let g be the constant map to a singleton. The composite is a bijection on a singleton, hence monic, but g is not injective.
Check your argument. Prove the implication and supply a fully typed counterexample.

MM7030-U1.07 · C · 30 min
Learning objective: U1.2 Split arrows
Show a morphism with a left inverse is monic. Give a monomorphism of sets with no left inverse.
Decisive idea. The map from the empty set to a singleton is injective.
If $rf = 1$, then $fu = fv$ implies $u = rfu = rfv = v$. The unique map $\varnothing \to 1$ is injective and therefore monic, but it has no left inverse because no map $1 \to \varnothing$ exists.
Check your argument. A left inverse must be an actual arrow with the required source.

MM7030-U1.08 · C · 30 min
Learning objective: U1.2 Epimorphisms
Prove a function missing an element of its codomain is not epic in sets.
Decisive idea. Make them differ only at the missing element.
If $f : X \to Y$ misses y_0, let $u : Y \to \{0, 1\}$ be constantly zero and let v be one at y_0 and zero elsewhere. Then $u \neq v$ but $uf = vf$, because f never reaches y_0. Hence f fails the epi cancellation property.
Check your argument. Specify the test maps and why their composites agree.

MM7030-U1.09 · C · 30 min
Learning objective: U1.1 Groupoids

Show the isomorphisms in any category, retaining all objects, form a subcategory in which every arrow is invertible.

Decisive idea. The inverse of gf is $f^{-1}g^{-1}$.

Identities are isomorphisms. For isomorphisms $f : A \to B$ and $g : B \to C$, the composite gf has proposed inverse $f^{-1}g^{-1}$. It satisfies both inverse equations by associativity, so composites of isomorphisms remain isomorphisms. The inverses themselves are isomorphisms, hence lie in this subcategory. It is a groupoid.

Check your argument. Verify the composition order.

MM7030-U1.10 · A · 50 min
Learning objective: U1.2 Balancedness

Prove a morphism that is both monic and split epic is an isomorphism.

Decisive idea. Compare $f(sf)$ with f.

Let $f : A \to B$ and choose its right inverse $s : B \to A$. Since $fs = 1_B$, we have $f(sf) = (fs)f = f = f1_A$. Monicity of f gives $sf = 1_A$. Thus s is both a right and left inverse, and f is an isomorphism.

Check your argument. Do not assume an arbitrary epi has a section.

MM7030-U1.11 · A · 50 min
Learning objective: U1.2 Rings versus sets

Prove the inclusion $\mathbb{Z} \to \mathbb{Z}\left[\frac{1}{2}\right]$ is epic among commutative unital rings but is not an isomorphism.

Decisive idea. Inverses, when they exist, are unique.

If two homomorphisms out of $\mathbb{Z}\left[\frac{1}{2}\right]$ agree on $\mathbb{Z}$, their values on $\frac{1}{2}$ are both inverses of the common image of two, hence agree. Every element is $\frac{a}{2^n}$, so all their values agree. This proves the inclusion epic. It is not onto because $\frac{1}{2}$ is not an integer, and a ring isomorphism must be a bijection.

Check your argument. Quantify over every possible target ring and compare all fractions.

MM7030-U1.12 · A · 50 min
Learning objective: U1.1 Categories from paths

Construct the free category on a directed graph and prove any assignment of its vertices and edges to objects and arrows of a category extends uniquely to a functor.

Decisive idea. A functor must send a path to the composite of its edge images.

Objects are vertices; arrows are finite composable paths; empty paths are identities; composition concatenates paths. Concatenation is associative. Given a graph assignment preserving endpoints, send each path to the composite of its edge images and each empty path to an identity. Associativity and identity laws in the target prove this is a functor. Every path is a composite of edges, so any extension must have these values, proving uniqueness.

Check your argument. Include empty paths and explain why no additional path relations are imposed.

2 Unit 2 · Functors and equivalences

MM7030-U2.01 · E · 15 min
Learning objective: U2.1 Functor laws

Show a monotone map between posets gives a functor between their categories.
Decisive idea. Monotonicity supplies the required arrow images.
If $a \leq b$, monotonicity gives $F(a) \leq F(b)$, so the unique arrow maps to the unique arrow. Identities and composites are preserved because the relevant hom sets contain at most one arrow.
Check your argument. Explain why monotonicity is exactly the condition needed.

MM7030-U2.02 · E · 15 min
Learning objective: U2.1 Forgetful functors

Define the forgetful functor from vector spaces over K to sets on objects and arrows.
Decisive idea. Linear maps are functions.
It sends a vector space to its underlying set and a linear map to its underlying function. Identity linear maps are identity functions and linear-map composition is function composition, proving the functor laws.
Check your argument. Describe both assignments.

MM7030-U2.03 · E · 15 min
Learning objective: U2.1 Hom functors

For $f : X \to Y$, write the map $\mathcal{C}(A, f)$.
Decisive idea. Its input is $h : A \to X$.
It is the function $\mathcal{C}(A, X) \to \mathcal{C}(A, Y)$, $h \mapsto fh$. The types ensure the composite exists.
Check your argument. Give domain, codomain, and formula.

MM7030-U2.04 · E · 15 min
Learning objective: U2.2 Faithfulness

Explain why the forgetful functor from groups to sets is faithful.
Decisive idea. Morphisms are actual functions with a property.
If two group homomorphisms have equal underlying functions, they take the same value at every group element and are the same homomorphism. Thus each induced hom-set map is injective.
Check your argument. Use the hom-set definition of faithful.

MM7030-U2.05 · C · 30 min
Learning objective: U2.1 Contravariance
Verify inverse image of subsets is a contravariant functor on sets.
Decisive idea. An element belongs to an inverse image exactly when its image belongs to the target subset.
For $f : X \to Y, g : Y \to Z$ and $S \subseteq Z$, $x \in (gf)^{-1}S$ exactly when $g(f(x)) \in S$, equivalently $x \in f^{-1}(g^{-1}S)$. Thus composition reverses. Identity inverse image is the identity on the power set.
Check your argument. State the reversed direction of the arrow map.

MM7030-U2.06 · C · 30 min
Learning objective: U2.2 Fullness
Show the forgetful functor from groups to sets is not full.
Decisive idea. Use a constant nonidentity value in a nontrivial group.
The constant function $C_2 \to C_2$ with value its nonidentity element is a set function but cannot be a group homomorphism, since homomorphisms send identity to identity. Therefore the hom-set map is not surjective.
Check your argument. Use a map between underlying sets of actual objects in the source category.

MM7030-U2.07 · C · 30 min
Learning objective: U2.2 Reflection of isomorphisms
Prove a full and faithful functor reflects isomorphisms.
Decisive idea. Use faithfulness on the two inverse equations.
If Ff has inverse u, fullness gives g with $Fg = u$. Then $F(gf) = FgFf = 1 = F(1)$ and similarly $F(fg) = F(1)$. Faithfulness gives $gf = 1$ and $fg = 1$, so f is an isomorphism.
Check your argument. Specify which property supplies the inverse candidate and which verifies its equations.

MM7030-U2.08 · C · 30 min
Learning objective: U2.1 One-object functors
Show functors between one-object categories of monoids are precisely monoid homomorphisms.
Decisive idea. Identity and composition become unit and multiplication.
The unique object must map to the unique object. The arrow map is a function between monoids. The two functor laws require it to preserve the unit and multiplication, exactly the monoid-homomorphism axioms. Conversely those axioms define a functor.
Check your argument. State both directions of the identification.

MM7030-U2.09 · C · 30 min
Learning objective: U2.2 Matrix category

Show the category of matrices over K, with objects nonnegative integers, is equivalent to finite-dimensional vector spaces.

Decisive idea. Every linear map between coordinate spaces has a unique matrix.

The matrix-to-linear-map functor preserves multiplication and identities. The uniqueness and existence of coordinate matrices make it full and faithful. Every finite-dimensional vector space has a finite basis and is isomorphic to some K^n, giving essential surjectivity. The equivalence criterion applies.

Check your argument. Include dimension zero and distinguish essential from literal surjectivity.

MM7030-U2.10 · A · 50 min
Learning objective: U2.2 Faithful reflection

Prove a faithful functor reflects monomorphisms and epimorphisms, without assuming fullness.

Decisive idea. Cancel in the target, then use faithfulness.

If Ff is monic and $fu = fv$, then $FfFu = FfFv$, so $Fu = Fv$ and faithfulness gives $u = v$. Thus f is monic. Dually, if Ff is epic and $uf = vf$, cancel Ff on the right and use faithfulness to conclude $u = v$.

Check your argument. Do not confuse reflection with preservation; fullness is unnecessary here.

MM7030-U2.11 · A · 50 min
Learning objective: U2.2 Equivalence of posets

Show equivalent poset categories are isomorphic as posets.

Decisive idea. A quasi-inverse is actually inverse on objects.

If F, G witness an equivalence, the component isomorphisms $GF(p) \cong p$ and $FG(q) \cong q$ force literal equality by antisymmetry. Thus the monotone maps are inverse bijections. Each direction is monotone, so they are an order isomorphism.

Check your argument. Use antisymmetry; the statement need not hold for preorders.

MM7030-U2.12 · A · 50 min
Learning objective: U2.2 Failure of preservation

Construct a faithful functor which sends a monomorphism and epimorphism to a function which is neither. Use the two-object poset $0 < 1$ as source, and explain why this does not contradict U2.10.

Decisive idea. Send the sole nonidentity arrow to a constant map between two-element sets.

Map both objects to the set $\{a, b\}$, identities to identities, and $0 \to 1$ to the constant function with value a. These assignments form a functor because every nontrivial composition involves an identity. It is faithful: each source hom set is empty or a singleton, so its map into the corresponding function set is injective. The nonidentity source arrow is both monic and epic, but its image is neither injective nor surjective, hence neither monic nor epic in sets. Faithfulness reflects cancellation properties from the target; it does not preserve them from the source.

Check your argument. Verify functoriality and faithfulness, and distinguish preservation from reflection.

3 Unit 3 · Natural transformations

MM7030-U3.01 · E · 15 min
Learning objective: U3.1 Naturality

Write the naturality equation for $\eta : F \Rightarrow G$ at $f : A \to B$.

Decisive idea. Both paths must run from FA to GB.

The equation is $G(f)\eta_A = \eta_B F(f)$. The left path goes first across at A and then down by Gf; the right path first uses Ff and then the component at B.

Check your argument. Check the domain and codomain of each factor.

MM7030-U3.02 · E · 15 min
Learning objective: U3.2 Identities

Show the identity maps 1_{FA} form a natural transformation $F \Rightarrow F$.

Decisive idea. Both sides reduce to Ff.

$Ff1_{FA} = Ff = 1_{FB}Ff$ by identity laws. Thus the component family is natural.

Check your argument. Use identity arrows at the correct objects.

MM7030-U3.03 · E · 15 min
Learning objective: U3.1 Natural projections

Show $\pi_1 : X \times X \to X$ is natural in the set X.

Decisive idea. Evaluate both composites on (x, y).

Both $f\pi_1$ and $\pi_1(f \times f)$ send (x, y) to $f(x)$. Hence the square commutes for every function f, proving naturality.

Check your argument. Quantify over all functions.

MM7030-U3.04 · E · 15 min
Learning objective: U3.2 Components

Describe a natural transformation between two constant functors with values A, B when the source is the terminal one-object category.

Decisive idea. The only naturality equation involves identity arrows.

It is any arrow $A \to B$. The sole naturality equation is $1_B\eta = \eta 1_A$, automatically true.

Check your argument. Explain why no additional compatibility condition occurs.

MM7030-U3.05 · C · 30 min
Learning objective: U3.1 Diagonal
Show the diagonal $\Delta_X : X \to X \times X$ is natural and each component is injective.
Decisive idea. Both square paths produce $(f(x), f(x))$.
For any $f : X \to Y$, $(f \times f)\Delta_{X(x)} = (f(x), f(x)) = \Delta_Y f(x)$, proving naturality. If $\Delta_{X(x)} = \Delta_{X(y)}$, equality of first coordinates gives $x = y$, so each component is injective.
Check your argument. Prove naturality separately from injectivity.

MM7030-U3.06 · C · 30 min
Learning objective: U3.2 Vertical composition
Prove vertical composition of natural transformations is associative.
Decisive idea. Use associativity in the target category.
For composable η, θ, κ, the components of $(\kappa\theta)\eta$ and $\kappa(\theta\eta)$ are $(\kappa_A\theta_A)\eta_A$ and $\kappa_{A(\theta_A\eta_A)}$, equal by associativity. Transformations are equal when all components agree, so composition is associative.
Check your argument. Include the componentwise equality criterion.

MM7030-U3.07 · C · 30 min
Learning objective: U3.1 Inverse naturality
If $\eta : F \Rightarrow G$ is natural and each component is invertible, prove the inverse family is natural.
Decisive idea. Start from $Gf\eta_A = \eta_B Ff$.
Compose on the left with η_B^{-1} and on the right with η_A^{-1} to obtain $\eta_B^{-1}Gf = Ff\eta_A^{-1}$. This is the naturality equation for a transformation $G \Rightarrow F$. Componentwise inverse equations make it the inverse transformation.
Check your argument. Keep the indices consistent.

MM7030-U3.08 · C · 30 min
Learning objective: U3.2 Group actions
For two actions of a group G on sets X, Y, show natural transformations of the corresponding functors are exactly equivariant functions.
Decisive idea. Naturality at g is the equivariance equation.
Because the group category has one object, the component family is one function T. Naturality at each arrow g says $T(g \cdot x) = g \cdot T(x)$ for every x. These are precisely the equations defining an equivariant function. Conversely an equivariant function satisfies every naturality square.
Check your argument. State both directions.

MM7030-U3.09 · C · 30 min
Learning objective: U3.1 Double dual

Verify naturality of the canonical map $V \to V^{**}$ by evaluating on a linear functional.
Decisive idea. For $f : V \to W$ and $\psi \in W^*$, both values are $\psi(f(v))$.
Let $\eta_{V(v)}(\varphi) = \varphi(v)$. Then $f^{**}\left(\eta_{V(v)}\right)(\psi) = \eta_{V(v)}(\psi f) = \psi(f(v)) = \eta_{W(f(v))}(\psi)$. Equality for every ψ gives $f^{**}\eta_V = \eta_W f$.
Check your argument. State the types of v, ψ, and the dual maps.

MM7030-U3.10 · A · 50 min
Learning objective: U3.1 Natural endomorphisms of sets

Classify all natural transformations from the identity functor on sets to itself without using Yoneda.
Decisive idea. The component on a singleton is forced to be its identity.
Let η be natural. For $x \in X$, let $i_x : 1 \to X$ select x. The only function $1 \to 1$ is the identity, so naturality gives $\eta_{X(x)} = \eta_X i_{x(*)} = i_x \eta_1(*) = x$. Thus every component is the identity, including the unique map on the empty set. The identity family is natural, so it is the only one.
Check your argument. Handle arbitrary elements via naturality, not a choice of ordering.

MM7030-U3.11 · A · 50 min
Learning objective: U3.2 Center of a group

Identify the natural endomorphisms of the identity functor of a group regarded as a one-object category.
Decisive idea. Naturality says it commutes with every group arrow.
A transformation is an element $z \in G$ with $gz = zg$ for every g. Thus its elements are exactly $Z(G)$. Vertical composition is multiplication of these components, and every component is invertible, so these transformations form the center as a group.
Check your argument. Describe both the set of transformations and its composition.

MM7030-U3.12 · A · 50 min
Learning objective: U3.1 Scalar natural operations

Show every natural endomorphism of the identity functor on K-vector spaces is multiplication by a single scalar.
Decisive idea. Use the linear map $K \to V$ taking one to a chosen vector.
Let $c = \eta_{K(1)}$. For $v \in V$, the map $i_v : K \to V$, $a \mapsto av$, is linear. Naturality gives $\eta_{V(v)} = \eta_V i_{v(1)} = i_v \eta_{K(1)} = cv$. Thus all components use the same scalar. Conversely scalar multiplication commutes with every linear map, so every $c \in K$ yields such a transformation.
Check your argument. Prove the scalar is independent of both the vector and the space.

4 Unit 4 · Universal properties

MM7030-U4.01 · E · 15 min
Learning objective: U4.1 Initial and terminal

Identify initial and terminal objects in sets and in groups.
Decisive idea. In groups, the trivial group serves both purposes.
In sets, the empty set is initial and any singleton is terminal. In groups, the trivial group is initial because its identity must map to the identity, and terminal because there is one homomorphism sending every element to the identity.
Check your argument. Explain uniqueness, not only existence.

MM7030-U4.02 · E · 15 min
Learning objective: U4.1 Products

Describe the product of a two-element set and a three-element set with its projections.
Decisive idea. The projections extract the two coordinates.
The Cartesian product has six pairs. The maps $(a, b) \mapsto a$ and $(a, b) \mapsto b$ are its projections. For maps $f : X \to A, g : X \to B$, the unique mediator is $x \mapsto (f(x), g(x))$.
Check your argument. Include the universal property.

MM7030-U4.03 · E · 15 min
Learning objective: U4.2 Coproducts

Why is ordinary union not always a coproduct of two sets?
Decisive idea. The two copies must be allowed to receive different images.
If $A = B = \{*\}$, ordinary union is a singleton. Two maps from A, B into a two-element target can send their elements to different values, and no map from the singleton union can extend both. A tagged disjoint union has two elements and does satisfy the coproduct property.
Check your argument. Use incompatible maps to demonstrate failure.

MM7030-U4.04 · E · 15 min
Learning objective: U4.1 Poset products

Find the product and coproduct of twelve and eighteen in positive integers ordered by divisibility.
Decisive idea. Use gcd and lcm.
The product is six and the coproduct is thirty-six. The universal properties are exactly: common divisors divide the gcd, and the lcm divides every common multiple.
Check your argument. Translate the arrow direction into divisibility.

MM7030-U4.05 · C · 30 min
Learning objective: U4.1 Product uniqueness
Prove two products of fixed objects are uniquely isomorphic compatibly with their projections.
Decisive idea. Their composites have the same projections as the identities.
The projections of each product provide a competing pair of maps for the other, yielding unique comparison maps. Each composite preserves both original projections, as does the identity. The uniqueness clause forces these composites to be identities. The comparison map is unique because its projections were prescribed.
Check your argument. Compatibility with projections is essential to the uniqueness claim.

MM7030-U4.06 · C · 30 min
Learning objective: U4.2 Finite biproducts
Prove $V \oplus W$ is both a product and a coproduct in vector spaces.
Decisive idea. The mediators are pairing and addition of images.
For a product, send x to $(f(x), g(x))$; linearity and uniqueness follow coordinatewise. For a coproduct, send (v, w) to $a(v) + b(w)$; this is linear and extends the given maps on the two summands. Every vector is $(v, 0) + (0, w)$, so this extension is unique.
Check your argument. Give both universal properties with their different arrow directions.

MM7030-U4.07 · C · 30 min
Learning objective: U4.2 Free vector spaces
Construct the linear extension of a function from $\{a, b\}$ into a vector space, taking a to u and b to v.
Decisive idea. A vector has unique coordinates $r[a] + s[b]$.
The unique extension sends $r[a] + s[b]$ to $ru + sv$. This formula is linear, has the specified basis values, and is forced by linearity for every input.
Check your argument. Explain uniqueness using the basis expansion.

MM7030-U4.08 · C · 30 min
Learning objective: U4.1 Terminal products
Show $A \times 1$ is isomorphic to A whenever 1 is terminal and the product exists.
Decisive idea. This pairing is inverse to the first projection.
Let $p : A \times 1 \to A$ and $s = \langle 1_A, ! \rangle : A \to A \times 1$. Then $ps = 1_A$. Both sp and the identity on the product have first projection p and the unique second projection to one; product uniqueness gives $sp = 1$. Thus they are inverse.
Check your argument. Use terminal uniqueness for the second-coordinate comparison.

MM7030-U4.09 · C · 30 min
Learning objective: U4.2 Groups and coproducts

Explain why $C_2 \times C_2$ with its coordinate inclusions is not a coproduct of two copies of C_2 in groups.
Decisive idea. Use (12) and (23).
Choose the involutions $(12), (23)$ in S_3. Each defines a homomorphism $C_2 \to S_3$. If a map from $C_2 \times C_2$ extended both, the two commuting coordinate generators would map to commuting elements. But $(12)(23) \neq (23)(12)$. Hence no such extension exists, violating the coproduct property.
Check your argument. Use a target and two valid homomorphisms for which extension fails.

MM7030-U4.10 · A · 50 min
Learning objective: U4.1 Product associativity

Construct the canonical isomorphism $(A \times B) \times C \to A \times (B \times C)$ solely from universal properties.
Decisive idea. Build nested pairings, then compare the coordinates of the two composites.
From the source, take its A, B, C coordinate maps a, b, c. Pair b, c to get a map to $B \times C$, then pair this with a. Construct the reverse map by pairing the first two coordinates. Each composite has the same three coordinate maps as the identity; two successive uses of product uniqueness force equality. Thus the maps are inverse and uniquely compatible with coordinates.
Check your argument. Avoid element notation unless specializing to sets.

MM7030-U4.11 · A · 50 min
Learning objective: U4.2 Infinite sums and products

For countably many copies of K, show the finite-support direct sum is a coproduct but generally not the product under its coordinate projections.
Decisive idea. A cone from K with every coordinate the identity would require an all-ones vector.
Given linear maps from each copy of K to V, send a finite-support tuple to the finite sum of its coordinate images. This defines the unique linear extension, proving the coproduct property. If the same object were a product, the cone from K with every coordinate identity would yield a map sending one to a tuple whose every coordinate is one. This tuple has infinite support and is absent, a contradiction.
Check your argument. Exhibit the cone whose mediator does not exist.

MM7030-U4.12 · A · 50 min
Learning objective: U4.1 Joint detection by projections

Prove product projections jointly determine every arrow into the product. Deduce the naturality of the canonical swap $A \times B \to B \times A$ in both variables.
Decisive idea. Both naturality-square composites have the same two projections.
If two arrows into a product have identical projection composites, both mediate the same pair of arrows, so they are equal. The swap is the pairing of the second and first projections. For maps $f : A \to A'$, $g : B \to B'$, the composites swap$(f \times g)$ and $(g \times f)$swap both have projections $g\pi_B$ and $f\pi_A$. Joint detection makes them equal, proving naturality.
Check your argument. Describe both projected composites with their types.

5 Unit 5 · Limits and colimits

MM7030-U5.01 · E · 15 min
Learning objective: U5.1 Equalizers

Find the equalizer in sets of $f, g : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n^2, g(n) = n$.

Decisive idea. The equation is $n(n-1) = 0$.

The equalizer is the inclusion $\{0, 1\} \to \mathbb{Z}$. Any function whose composites with f, g agree has all values in this subset and factors uniquely through its inclusion.

Check your argument. Give the structural map and universal factorization.

MM7030-U5.02 · E · 15 min
Learning objective: U5.1 Pullbacks

Find the pullback of $f : \mathbb{Z} \to \mathbb{Z}$, $f(a) = 2a$, and $g : \mathbb{Z} \to \mathbb{Z}$, $g(b) = 3b$, in sets.

Decisive idea. Coprimality implies $a = 3k, b = 2k$.

The pullback is the set $\{(3k, 2k) : k \in \mathbb{Z}\}$ with coordinate projections. It can be identified with $\mathbb{Z}$ by $k \mapsto (3k, 2k)$. Compatibility of any pair of functions forces each value to be such a pair, giving the unique mediator.

Check your argument. Distinguish the abstract apex from its two projection maps.

MM7030-U5.03 · E · 15 min
Learning objective: U5.2 Coequalizers

In sets, coequalize the two maps from a singleton to $\{a, b, c\}$ selecting a and b.

Decisive idea. The equivalence classes are $\{a, b\}$ and $\{c\}$.

The coequalizer is the quotient onto the classes $\{a, b\}$ and $\{c\}$. A map from $\{a, b, c\}$ coequalizes the pair exactly when its values at a, b agree, exactly the condition for a unique map from this quotient.

Check your argument. Describe the factorization condition.

MM7030-U5.04 · E · 15 min
Learning objective: U5.1 Empty diagrams

Explain why a limit of the empty diagram is terminal and a colimit is initial.

Decisive idea. The mediator must therefore be unique for every object.

A cone over the empty diagram has no structural arrows or compatibility conditions. Its universal property asks for a unique arrow from each object to the chosen apex, which is terminality. Reversing arrows gives initiality for the colimit.

Check your argument. Explain why no extra cone data remain.

MM7030-U5.05 · C · 30 min
Learning objective: U5.1 Equalizers are monic
Prove an equalizer arrow is monic using only its universal property.
Decisive idea. They are mediators of the same equalizing arrow.
For equalizer $e : E \to A$, if $eu = ev$, this common composite equalizes the original pair. Both u, v factor it through e, so uniqueness of the factorization yields $u = v$.
Check your argument. Do not assume the category has elements.

MM7030-U5.06 · C · 30 min
Learning objective: U5.1 Pullbacks of monos
Prove a pullback of a monomorphism is monic.
Decisive idea. Cancel the original monomorphism to obtain equality of the other projected composites.
Let the pullback square have $p : P \to A, q : P \to B$, with $f : A \to C$ monic and $fp = gq$. If $qu = qv$, then $fpu = gqu = gqv = fpv$, so $pu = pv$. Both projected composites now agree, and pullback uniqueness gives $u = v$. Thus q is monic.
Check your argument. Identify which projection is the pullback of the given mono.

MM7030-U5.07 · C · 30 min
Learning objective: U5.2 Linear coequalizers
Prove the coequalizer of linear maps $f, g : V \to W$ is $W/\operatorname{im}(f - g)$.
Decisive idea. Use the universal property of a quotient vector space.
For $h : W \to X$, the condition $hf = hg$ is equivalent to $h(f - g) = 0$, hence to $\operatorname{im}(f - g) \subseteq \ker h$. This is exactly the condition for a unique linear factor through the quotient map. Thus the quotient has the coequalizer property.
Check your argument. Prove the equivalence of the equations and the kernel condition.

MM7030-U5.08 · C · 30 min
Learning objective: U5.2 Pushout of sets
Compute the pushout of inclusions $\{b\} \to \{a, b\}$ and $\{b\} \to \{b, c\}$ in sets.
Decisive idea. Identify the two copies of b.
The resulting set has three elements represented by a, b, c. The two inclusion maps agree on the common b. Any pair of maps from the original sets agreeing at b determines one function on these three elements and conversely, proving universality.
Check your argument. Begin with distinct tagged copies before imposing the identification.

MM7030-U5.09 · C · 30 min
Learning objective: U5.1 Limits in groups
Show the equalizer of group homomorphisms $f, g : G \to H$ is a subgroup of G. Must it be normal?
Decisive idea. For failure of normality compare an inner automorphism with the identity.
If $f(a) = g(a)$ and $f(b) = g(b)$, then $f(ab^{-1}) = f(a)f(b)^{-1} = g(a)g(b)^{-1} = g(ab^{-1})$, proving subgroup closure. It need not be normal: in S_3, the equalizer of conjugation by (12) and the identity is its centralizer $\{e, (12)\}$, which is not normal.
Check your argument. Distinguish a general equalizer from a homomorphism kernel.

MM7030-U5.10 · A · 50 min
Learning objective: U5.1 Finite-limit construction
Construct the pullback of $f : A \to C, g : B \to C$ using a product and an equalizer.
Decisive idea. Equalize $f\pi_A$ and $g\pi_B$.
Let $e : P \to A \times B$ equalize $f\pi_A, g\pi_B : A \times B \to C$. The composites $\pi_A e, \pi_B e$ form a compatible square. A compatible pair $u : X \to A, v : X \to B$ gives a unique product map $\langle u, v\rangle$ that equalizes the two maps, hence factors uniquely through e. This is precisely the pullback property.
Check your argument. Use both universal properties in sequence and track uniqueness.

MM7030-U5.11 · A · 50 min
Learning objective: U5.2 Gluing may identify more
Give a coequalizer in sets where the initially required identification pairs are not themselves transitive, and compute the actual quotient.
Decisive idea. Require $a \sim b$ and $b \sim c$.
Take $A = \{0, 1\}$, $B = \{a, b, c\}$, with $f(0) = a, g(0) = b$ and $f(1) = b, g(1) = c$. The listed pairs do not include $a \sim c$, but any coequalizing function has $h(a) = h(b) = h(c)$. Thus all three elements must be identified, giving a singleton quotient.
Check your argument. Explain why transitive closure is forced by equality in every target.

MM7030-U5.12 · A · 50 min
Learning objective: U5.1 Pullback pasting
Prove that two horizontally composable pullback squares have a pullback as their outer rectangle.
Decisive idea. Then use the left square; reverse this reasoning for uniqueness.
Given maps to the outer lower-left and upper-right corners whose composites to the lower-right agree, compose the first map to the lower-middle corner. The right pullback gives a unique map to the upper-middle corner. Its compatibility with the lower-left map is now exactly the hypothesis for the left pullback, yielding a map to the upper-left corner. Any outer mediator must give the same middle map by right-square uniqueness and then the same left map by left-square uniqueness. Thus the outer square is a pullback.
Check your argument. Specify the order of the two factorizations and both uniqueness steps.

6 Unit 6 · Representability and Yoneda

MM7030-U6.01 · E · 15 min
Learning objective: U6.1 Yoneda evaluation

For $\eta : \mathcal{C}(A, -) \Rightarrow F$, which element determines η under Yoneda?
Decisive idea. Use its identity arrow.
It is $\eta_{A(1_A)} \in FA$. Naturality reconstructs each value by $\eta_{B(f)} = F(f)\big(\eta_{A(1_A)}\big)$.
Check your argument. Include the reconstruction formula.

MM7030-U6.02 · E · 15 min
Learning objective: U6.2 Representing the identity

Show the identity functor on sets is covariantly represented by a singleton.
Decisive idea. Evaluate at its unique point.
The bijection $\mathrm{Set}(1, X) \to X$ sends f to $f(*)$. Its inverse sends x to the selecting function. Postcomposing with a function $X \to Y$ corresponds to applying it to the selected element, so the bijection is natural.
Check your argument. Verify naturality, not only cardinality.

MM7030-U6.03 · E · 15 min
Learning objective: U6.1 Reconstructing a transformation

Let $x \in FA$. Define the corresponding transformation from $\mathcal{C}(A, -)$ to F.
Decisive idea. The component at B sends f to $F(f)(x)$.
Define $\eta_{B(f)} = F(f)(x)$. For $g : B \to C$, $F(g)\eta_{B(f)} = F(gf)(x) = \eta_{C(gf)}$, so the family is natural.
Check your argument. Give its formula and naturality check.

MM7030-U6.04 · E · 15 min
Learning objective: U6.2 Universal elements

For the underlying-set functor on groups represented by $\mathbb{Z}$, identify a universal element.
Decisive idea. Choose the integer one.
The universal element is $1 \in \mathbb{Z}$. Every $g \in G$ is the image of one under the unique homomorphism $n \mapsto g^n$. This is exactly the universal-element property.
Check your argument. State the unique homomorphism explicitly.

MM7030-U6.05 · C · 30 min
Learning objective: U6.1 Natural binary operations

Classify natural transformations $X \times X \to X$ in sets.

Decisive idea. Yoneda gives one transformation for each element of that set.

The functor $X \times X$ is naturally $\mathrm{Set}(2, X)$. Yoneda identifies its transformations to the identity with the two elements of 2. Reconstruction evaluates a map $2 \to X$ at the selected element, giving exactly the two projections.

Check your argument. Explain the representable identification and reconstructed operations.

MM7030-U6.06 · C · 30 min
Learning objective: U6.2 Contravariant representables

Given $f : A \to B$, write the natural transformation $\mathcal{C}(-, A) \Rightarrow \mathcal{C}(-, B)$ and prove naturality.

Decisive idea. Precomposition with a test arrow commutes by associativity.

At X, send $u : X \to A$ to $fu : X \to B$. For $k : Y \to X$, $(fu)k = f(uk)$, so the square with the contravariant hom-functor maps commutes.

Check your argument. Keep the test-arrow direction explicit.

MM7030-U6.07 · C · 30 min
Learning objective: U6.2 Universal element category

Show a universal element of a covariant functor is an initial object in its category of elements.

Decisive idea. Such an arrow is exactly a map satisfying $F(f)(u) = x$.

From (A, u) to (X, x), the category of elements permits precisely arrows $f : A \to X$ with $F(f)(u) = x$. Universality gives exactly one such arrow for every target object, which is the definition of initiality. The same unpacking proves the converse.

Check your argument. State both directions.

MM7030-U6.08 · C · 30 min
Learning objective: U6.1 Empty representing set

Classify natural transformations from the constant singleton functor on sets to the identity functor.

Decisive idea. Yoneda asks for an element of the empty set.

Since $\mathrm{Set}(\varnothing, X)$ is always a singleton, Yoneda identifies the transformations with elements of $\varnothing$, so there are none. Directly, the component at the empty set would have to be a function $1 \to \varnothing$, which does not exist.

Check your argument. Give either full argument and recognize the empty-set obstruction.

MM7030-U6.09 · C · 30 min
Learning objective: U6.2 Uniqueness

Prove that naturally isomorphic contravariant representables have isomorphic representing objects.
Decisive idea. Faithfulness translates their composite identities back to arrows.
If $h_A \cong h_B$, the natural isomorphism and its inverse correspond to arrows $f : A \to B, g : B \to A$. Their composites correspond to identity transformations. Faithfulness of the embedding gives $gf = 1_A$ and $fg = 1_B$, so $A \cong B$.
Check your argument. Explain why the arrows are inverse, not merely that both exist.

MM7030-U6.10 · A · 50 min
Learning objective: U6.1 Natural power operations on groups

Classify all natural transformations from the underlying-set functor on groups to itself.
Decisive idea. An integer n produces the operation $g \mapsto g^n$.
Yoneda identifies these transformations with the underlying set of $\mathbb{Z}$, so each corresponds to one integer n. For $g \in G$, use the homomorphism $f_g : \mathbb{Z} \to G$ sending one to g. Reconstruction gives $\eta_{G(g)} = f_{g(n)} = g^n$. Every such power operation is natural because homomorphisms preserve integer powers. The component functions need not themselves be homomorphisms.
Check your argument. State the last distinction; the target category of the functor is sets.

MM7030-U6.11 · A · 50 min
Learning objective: U6.2 Nonrepresentability

Show the constant empty-set functor from sets to sets is not covariantly representable.
Decisive idea. Every object has an identity arrow.
If it were represented by A, then $\mathrm{Set}(A, A)$ would be in bijection with the empty set. But it contains 1_A, even when A is empty. This contradiction proves nonrepresentability.
Check your argument. Use the identity argument for every possible representing set.

MM7030-U6.12 · A · 50 min
Learning objective: U6.2 Natural n-ary operations

For finite sets m, n, classify natural transformations $X^n \to X^m$ on sets, including the empty cases.
Decisive idea. Yoneda identifies transformations with functions $m \to n$.
Yoneda gives $\mathrm{Nat}(\mathrm{Set}(n, -), \mathrm{Set}(m, -)) \cong \mathrm{Set}(m, n)$. A function $s : m \to n$ sends an n-tuple x to the m-tuple $j \mapsto x(s(j))$, so all operations select and repeat coordinates. If m is empty there is one transformation; if n is empty and m is nonempty there are none. If both are empty there is one.
Check your argument. Explain the reversed direction and all empty-set cases.

7 Unit 7 · Adjunctions

MM7030-U7.01 · E · 15 min
Learning objective: U7.1 Adjunction types

For $F : \mathcal{C} \to \mathcal{D}$ left adjoint to G, write the hom-set bijection and the types of the unit and counit components.

Decisive idea. The unit starts at C, the counit ends at D.

The bijection is $\mathcal{D}(FC, D) \cong \mathcal{C}(C, GD)$. The unit has type $\eta_C : C \to GFC$ and the counit $\varepsilon_D : FGD \to D$.

Check your argument. Check every object belongs to the category of the displayed hom set.

MM7030-U7.02 · E · 15 min
Learning objective: U7.1 Currying

Curry the function $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$, $f(b, a) = 2b - a$.

Decisive idea. The result is a function assigning a function.

The transpose sends b to the function $a \mapsto 2b - a$. Uncurrying evaluates that function at a, recovering the original formula.

Check your argument. State both the curried type and formula.

MM7030-U7.03 · E · 15 min
Learning objective: U7.2 Evaluation

Describe the counit of the product–exponential adjunction in sets.

Decisive idea. The domain is $X^A \times A$.

The counit is $\varepsilon_{X(f,a)} = f(a)$ from $X^A \times A$ to X. It is the uncurried identity on X^A.

Check your argument. Identify the adjunct correspondence producing it.

MM7030-U7.04 · E · 15 min
Learning objective: U7.2 Free-vector-space unit

Describe the unit of the free-vector-space/forgetful adjunction.

Decisive idea. The source is a set, not a vector space.

At a set X, the unit is $x \mapsto [x]$ from X to the underlying set of the free vector space K^X. It is a set function, not a purported linear map on X.

Check your argument. Keep the source category clear.

MM7030-U7.05 · C · 30 min
Learning objective: U7.1 Power-set adjunction

Prove direct image along $f : X \to Y$ is left adjoint to inverse image on power-set posets.

Decisive idea. The condition is $f[S] \subseteq T$ iff $S \subseteq f^{-1}[T]$.

We need $f[S] \subseteq T$ exactly when $S \subseteq f^{-1}[T]$. The left inclusion says every $s \in S$ has $f(s) \in T$. This is exactly that every $s \in S$ belongs to $f^{-1}[T]$, the right inclusion. Both operations are monotone, and the equivalence gives the poset adjunction.

Check your argument. Check monotonicity and the two-way inclusion equivalence.

MM7030-U7.06 · C · 30 min
Learning objective: U7.1 Floor and inclusion

Let $i : \mathbb{Z} \to \mathbb{R}$ be inclusion as ordered sets. Show i is left adjoint to floor.

Decisive idea. For integer n, $n \leq x$ iff $n \leq \lfloor x \rfloor$.

Floor is monotone. Since $\lfloor x \rfloor$ is the greatest integer at most x, an integer n satisfies $i(n) \leq x$ exactly when $n \leq \lfloor x \rfloor$. This is the adjunction criterion for posets.

Check your argument. Identify which map is left adjoint; ceiling gives the other adjunction.

MM7030-U7.07 · C · 30 min
Learning objective: U7.2 Free counit

For a vector space V, describe the counit $FUV \to V$ and verify the triangle involving the unit on UV.

Decisive idea. A single formal basis vector $[v]$ evaluates to v.

The counit sends $\sum_i a_{i[v_i]}$ to $\sum_i a_i v_i$. The unit on the underlying set sends v to $[v]$. Their composite, after applying the forgetful functor as necessary, sends v to v, so $U(\varepsilon_V)\eta_{UV} = 1_{UV}$.

Check your argument. Distinguish formal sums from actual vector-space sums.

MM7030-U7.08 · C · 30 min
Learning objective: U7.2 Transpose formulas

Given unit and counit, write the transpose of $h : FC \to D$ and the inverse transpose of $k : C \to GD$.

Decisive idea. The formulas are $Gh\eta_C$ and $\varepsilon_D Fk$.

The transpose is $G(h)\eta_C : C \to GD$. The inverse transpose is $\varepsilon_D F(k) : FC \to D$. Their domains match because the unit lands in GFC and $F(k)$ lands in FGD. The triangle identities make the two procedures inverse.

Check your argument. Check the intermediate objects.

MM7030-U7.09 · C · 30 min
Learning objective: U7.1 Diagonal adjunctions

Show the diagonal functor $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ is left adjoint to the binary-product functor when products exist.

Decisive idea. The product universal property supplies the corresponding map into $A \times B$.

The hom set in the product category is $\mathcal{C}(X, A) \times \mathcal{C}(X, B)$. The product property identifies this naturally with $\mathcal{C}(X, A \times B)$. This is exactly $\Delta \dashv \times$, with the product operation regarded as a functor defined by pairing.

Check your argument. Mention naturality of the pairing correspondence.

MM7030-U7.10 · A · 50 min
Learning objective: U7.2 Triangle converse

Prove the two unit–counit transpose formulas are inverse assuming naturality and the triangle identities.

Decisive idea. Move the map past the counit or unit using naturality.

For h, the round trip is $\varepsilon_D FG(h) F\eta_C = h\varepsilon_{FC} F\eta_C = h$. For k, it is $G\varepsilon_D GF(k)\eta_C = G\varepsilon_D \eta_{GD} k = k$. The middle equalities use counit and unit naturality; the final equalities use the corresponding triangle.

Check your argument. Name the naturality instance and triangle used in each direction.

MM7030-U7.11 · A · 50 min
Learning objective: U7.1 Composite adjunctions

If $F \dashv G$ and $F' \dashv G'$ are composable adjunctions, prove $F'F \dashv \mathbb{G}'$.

Decisive idea. Track the order of the right adjoints.

For appropriate objects C, E, the natural bijections are $\mathcal{E}(F'FC, E) \cong \mathcal{D}(FC, G'E) \cong \mathcal{C}(C, GG'E)$. Their composite is a bijection natural in both variables, so it is an adjunction. The right adjoints appear in reversed composition order.

Check your argument. Specify the three categories and ensure the functors compose.

MM7030-U7.12 · A · 50 min
Learning objective: U7.2 Closure operators

For an adjunction $F : P \to Q, G : Q \to P$ of posets, prove GF is monotone, extensive, and idempotent.

Decisive idea. Use the counit inequality on Fp for the reverse idempotence inequality.

Monotonicity follows by composition. The adjunction applied to $Fp \leq Fp$ gives $p \leq GFp$. Applying this to GFp gives $GFp \leq GFGFp$. The counit gives $FG(Fp) \leq Fp$; applying monotone G gives $GFGFp \leq GFp$. Antisymmetry yields equality, proving idempotence.

Check your argument. Use both inequalities and antisymmetry.

8 Unit 8 · Preservation and applications

MM7030-U8.01 · E · 15 min
Learning objective: U8.1 Preservation direction
Which of products and coproducts must a right adjoint preserve, when they exist?
Decisive idea. Products are limits.
A right adjoint preserves products because they are limits. No general coproduct-preservation statement follows from being a right adjoint alone.
Check your argument. State the existence qualification and avoid claiming a converse.

MM7030-U8.02 · E · 15 min
Learning objective: U8.1 Terminal objects
Explain why a right adjoint preserves terminal objects.
Decisive idea. Apply limit preservation.
A terminal object is the limit of the empty diagram, so its image under a right adjoint is terminal by the limit-preservation theorem. Equivalently its maps from X correspond by adjunction to the unique map from FX into the original terminal object.
Check your argument. Give one complete justification.

MM7030-U8.03 · E · 15 min
Learning objective: U8.2 Abelianization
What is the abelianization of an abelian group?
Decisive idea. The quotient is by the trivial subgroup.
For abelian A, every $aba^{-1}b^{-1} = e$, so the commutator subgroup is trivial and $A^{\mathrm{ab}} = A/\{e\}$ is canonically isomorphic to A.
Check your argument. Identify the quotient map under the isomorphism.

MM7030-U8.04 · E · 15 min
Learning objective: U8.2 Exponentials
In sets, how many functions are in B^A if $|A| = 3$ and $|B| = 2$? Explain its role in currying.
Decisive idea. It is the object representing maps out of a product with A.
There are $2^3 = 8$ functions. The natural bijection $\mathrm{Set}(X \times A, B) \cong \mathrm{Set}\big(X, B^A\big)$ sends a function of two arguments to its family of functions on A.
Check your argument. Include the representing property, not only the count.

MM7030-U8.05 · C · 30 min
Learning objective: U8.1 Representable preservation
Show $\mathcal{C}(A, -)$ preserves a binary product.
Decisive idea. A map into the product is equivalent to a pair of component maps.
The comparison map sends $h : A \to B \times C$ to $(\pi_B h, \pi_C h)$. Product universality supplies its inverse, pairing the two arrows. Thus it is a bijection respecting projections, which proves preservation of the product in sets.
Check your argument. Identify the actual comparison map.

MM7030-U8.06 · C · 30 min
Learning objective: U8.1 Nonpreservation
Show the free-vector-space functor does not preserve the terminal set.
Decisive idea. The terminal vector space has dimension zero.
The terminal set is a singleton; its free vector space is one-dimensional. The terminal vector space is zero-dimensional, so the image is not terminal. This does not contradict left-adjoint preservation, which concerns colimits rather than limits.
Check your argument. Explain the direction of the theorem.

MM7030-U8.07 · C · 30 min
Learning objective: U8.2 Abelianization map
Compute the abelianization of S_3.
Decisive idea. A commutator of two transpositions is a nontrivial three-cycle.
Sign kills commutators, so $[S_3, S_3] \subseteq A_3$. The commutator of (12) and (23) is a nontrivial three-cycle, which generates A_3. Thus equality holds and $S_3^{\mathrm{ab}} \cong S_3/A_3 \cong C_2$.
Check your argument. Prove both containments for the commutator subgroup.

MM7030-U8.08 · C · 30 min
Learning objective: U8.2 Free coproducts
Construct the canonical isomorphism $F(A \coprod B) \cong F(A) \oplus F(B)$ for free vector spaces.
Decisive idea. Use linear extension in both directions.
Map a basis vector from the A tag to $([a], 0)$ and from the B tag to $(0, [b])$, then extend linearly. The inverse sends each summand basis vector to its tagged basis vector. Both composites are identity on every basis vector, hence on the whole space.
Check your argument. Explain why equality on generators suffices.

MM7030-U8.09 · C · 30 min
Learning objective: U8.2 Implication in a poset

For subsets of S, prove $X \cap A \subseteq B$ iff $X \subseteq (S \setminus A) \cup B$.

Decisive idea. If it does, the first inclusion forces membership in B.

If the first inclusion holds and $x \in X$, then either $x \notin A$ or $x \in X \cap A \subseteq B$, proving the right inclusion. Conversely if $x \in X \cap A$, the right inclusion places it in the displayed union, and membership in A excludes its complement, so $x \in B$.

Check your argument. Prove both directions by element membership.

MM7030-U8.10 · A · 50 min
Learning objective: U8.1 Right-adjoint preservation

Prove directly through adjunct transposes that a right adjoint preserves an equalizer.

Decisive idea. Naturality transfers the equality to the original pair.

Let $e : E \to A$ equalize $f, g : A \to B$, and let $u : X \to GA$ satisfy $Gfu = Ggu$. Its transpose $v : FX \to A$ satisfies $fv = gv$ by naturality. There is a unique $w : FX \to E$ with $ew = v$. Transpose w to $z : X \to GE$; naturality gives $Gez = u$. Any second factor would transpose to a second factor through e, impossible. Thus Ge is the equalizer.

Check your argument. Include the uniqueness argument after transposing back.

MM7030-U8.11 · A · 50 min
Learning objective: U8.2 Abelianization adjunction

Prove abelianization is left adjoint to inclusion of abelian groups, including naturality.

Decisive idea. Use uniqueness of quotient factorization under pre- and postcomposition.

A homomorphism $f : G \to A$ with A abelian kills each commutator and hence factors uniquely through G^{ab}. Conversely composing a map from the quotient with its projection gives such an f, and the constructions are inverse. Precomposition by a group homomorphism respects commutators, while postcomposition by an abelian-group homomorphism respects factorization. In each case both routes give the unique factor of the same composite, proving naturality.

Check your argument. Establish the bijection and both variables' compatibility.

MM7030-U8.12 · A · 50 min
Learning objective: U8.1 Equivalences preserve limits

Prove an equivalence of categories preserves all existing small limits.

Decisive idea. Use fullness to lift its arrows and faithfulness to lift the cone equations.

Let F be full, faithful, and essentially surjective, and let L be a limit. A cone in the target with apex X can be transferred along an isomorphism $FA \cong X$. Fullness lifts its legs to arrows from A; faithfulness shows they satisfy the cone equations. Limit universality gives a unique arrow $A \to L$. Applying F and composing with the inverse chosen isomorphism gives a mediator from X. Fullness and faithfulness translate any rival mediator back to one in the source, proving uniqueness.

Check your argument. Explain the roles of all three equivalence properties.

9 Mock assessment 1

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M1.01 · E · 15 min

Learning objective: U1 Cancellation

State the definition of epimorphism and prove every surjective set function is epic.

Decisive idea. Evaluate at a preimage of each target element.

An arrow $e : A \to B$ is epic if $ue = ve$ implies $u = v$ for all parallel maps out of B. If e is a surjective function, choose for each b any a with $e(a) = b$. Then $u(b) = u(e(a)) = v(e(a)) = v(b)$. Thus $u = v$.

Check your argument. No simultaneous choice of preimages is needed; argue for an arbitrary b.

MM7030-M1.02 · E · 15 min

Learning objective: U4 Universal properties

State the coproduct property and identify the coproduct of two finite sets of sizes three and four.

Decisive idea. The resulting set has seven elements.

The coproduct is a tagged disjoint union of size seven. For maps from the two sets to any X, define the mediating function on each tagged component by the given map. It is unique because every element belongs to exactly one component.

Check your argument. Include the maps and uniqueness clause.

MM7030-M1.03 · C · 30 min

Learning objective: U3 Naturality

Prove the swap maps $X \times X \to X \times X$, $(x, y) \mapsto (y, x)$ form a natural isomorphism.

Decisive idea. Both square paths send (x, y) to $(f(y), f(x))$.

For any $f : X \to Y$, both $(f \times f)\mathrm{swap}_X$ and $\mathrm{swap}_{Y(f\times f)}$ send (x, y) to $(f(y), f(x))$, proving naturality. Each swap composed with itself is the identity, so each component is invertible; the inverse family is the same natural transformation. Thus it is a natural isomorphism.

Check your argument. Check naturality and invertibility independently.

MM7030-M1.04 · C · 30 min
Learning objective: U5 Limits

Construct the pullback in sets of two maps to a singleton and prove its universal property.

Decisive idea. The construction is a Cartesian product.

For $A \to 1$ and $B \to 1$, every pair (a, b) is compatible, so the pullback is $A \times B$. Given maps from X to A, B, the unique mediator is their pairing. Both composites to the singleton agree automatically, so this is exactly the pullback property.

Check your argument. Explain why there is no remaining compatibility restriction.

MM7030-M1.05 · A · 50 min
Learning objective: U6 Yoneda

Classify natural transformations $X^3 \to X^2$ in sets and count them.

Decisive idea. They correspond to functions from a two-element set to a three-element set.

Yoneda gives $\mathrm{Nat}(\mathrm{Set}(3, -), \mathrm{Set}(2, -)) \cong \mathrm{Set}(2, 3)$. Each of the two output coordinates independently selects one of the three input coordinates. There are $3^2 = 9$ transformations, explicitly $(x_1, x_2, x_3) \mapsto (x_i, x_j)$ for $i, j \in \{1, 2, 3\}$.

Check your argument. Justify exhaustiveness with Yoneda.

MM7030-M1.06 · A · 50 min
Learning objective: U7–U8 Adjunctions

Prove a left adjoint preserves initial objects directly from its hom-set bijection.

Decisive idea. The corresponding source hom set is a singleton.

If I is initial in $\mathcal{C}$ and $F \dashv G$, then for every D the adjunction gives $\mathcal{D}(FI, D) \cong \mathcal{C}(I, GD)$. The right set has exactly one element, so the left does too. This is precisely initiality of FI.

Check your argument. Use all targets D, not one particular object.

10 Mock assessment 2

Allow 180 minutes. Attempt all six problems and explain every inference. Use the rubric in the handbook for self-assessment; this is original practice, not an official examination.

MM7030-M2.01 · E · 15 min

Learning objective: U2 Functors

Prove every functor preserves isomorphisms.

Decisive idea. The image of an inverse is an inverse.

If $gf = 1$ and $fg = 1$, then functoriality gives $FgFf = F(gf) = 1$ and $FfFg = F(fg) = 1$. Thus Ff is an isomorphism with inverse Fg.

Check your argument. Use identity preservation as well as composition preservation.

MM7030-M2.02 · E · 15 min

Learning objective: U7 Unit and counit

For free vector spaces, describe the unit at a set X and counit at a vector space V.

Decisive idea. Insertion and evaluation are the two maps.

The unit sends x to its basis vector $[x]$ in the underlying set of FX. The counit sends $\sum_i a_{i[v_i]}$ in FUV to the actual vector sum $\sum_i a_i v_i$ in V.

Check your argument. Give the source and target of each map.

MM7030-M2.03 · C · 30 min

Learning objective: U5 Coequalizers

Compute the coequalizer in vector spaces of $f, g : K \to K^2$ given by $f(t) = (t, 0)$ and $g(t) = (0, t)$.

Decisive idea. The map $(a, b) \mapsto a + b$ has the required kernel.

The image of $f - g$ is the line spanned by $(1, -1)$. The quotient is one-dimensional and is identified with K by $(a, b) \mapsto a + b$, whose kernel is that line and which is surjective. This map coequalizes f, g and has the quotient universal property.

Check your argument. The formula also works in characteristic two.

MM7030-M2.04 · C · 30 min
Learning objective: U6 Representability

Show the functor $G \mapsto G \times G$ from groups to sets is represented by a free group on two generators.

Decisive idea. Any ordered pair of group elements is permitted.

Let F_2 be free on a, b. Sending a homomorphism to $(f(a), f(b))$ is bijective by the free-group property. Postcomposition applies a group homomorphism to both entries, so this bijection is natural. Hence the underlying-pair functor is represented by F_2.

Check your argument. Use the free-group universal property, not the direct product of two copies of $\mathbb{Z}$.

MM7030-M2.05 · A · 50 min
Learning objective: U2–U5 Equivalence

Let F be full, faithful, and essentially surjective. Prove it sends a terminal object to a terminal object.

Decisive idea. Use the hom-set bijection supplied by full faithfulness.

For any target object D, choose an isomorphism $FA \to D$. Maps $D \to FT$ correspond by precomposition to maps $FA \to FT$, and full faithfulness identifies the latter with maps $A \to T$. Since T is terminal, this set is a singleton. Thus every D has exactly one map to FT, proving terminality.

Check your argument. Explain why essential surjectivity is needed to cover every D.

MM7030-M2.06 · A · 50 min
Learning objective: U7–U8 Logic

In the power-set poset of S, identify a right adjoint to intersection with a fixed A and derive its counit inequality.

Decisive idea. The largest choice is $(S \setminus A) \cup B$.

The right adjoint is $G(B) = (S \setminus A) \cup B$, since $X \cap A \subseteq B$ iff $X \subseteq G(B)$. Its counit is $G(B) \cap A \subseteq B$, which holds because an element in A cannot be in its complement and must therefore belong to B.

Check your argument. State the adjunction equivalence and derive the correctly oriented inequality.